When the LLM Judge Silently Fails: A Context-Overflow Failure Mode in Faithful RAG Evaluation

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Abstract

We report a silent failure mode in LLM-as-judge evaluation of retrieval-augmented generation (RAG), found while running a controlled ablation of a local pipeline over 60 handwritten questions on 12 RAG papers. An LLM judge returns schema-valid JSON with every scored field set to zero when its input context exceeds ∼2,500 tokens in our setup, producing apparent pipeline collapses (∼10/100) that are evaluation artifacts rather than generation failures: JSON parse rate stays at 100% and deterministic citation accuracy near 80% throughout. After we cap the judge's context and re-score stored outputs, the affected conditions recover but still regress relative to baseline (−17.7 and −15.9 points for top-k=10 and 512-word chunks)—a real but far smaller effect that the artifact had masked. We add two supporting audit lessons from the same ablation: a naive post-reranking sentence filter costs 12.9 points when restricted to the first five sentences per chunk despite unchanged retrieval, and a case study where document-level Recall@5 stays 100% while PDF chunking splits the answer across a boundary. We release 66 ablation runs with pre- and post-fix judge traces; an anonymized supplementary archive accompanies this submission.

1 Introduction

Retrieval-augmented generation (RAG) pipelines combine interchangeable components—retrievers, rerankers, context filters, and generators—each with hyperparameters that affect whether a model can ground answers in retrieved evidence (Lewis et al., 2020; Gao et al., 2023). End-to-end benchmarks (Chen et al., 2024; Es et al., 2024) and best-practice studies (Wang et al., 2024; Li et al., 2025) report aggregate scores that depend on the evaluation stack—often an LLM-as-judge—being trustworthy. We show that this assumption can fail silently: an LLM judge can emit schema-valid, all-zero scores that look like genuine grounding failures but are artifacts of judge input length, not model behavior.

We found this while running single-variable ablations of a fully local RAG pipeline (Qdrant, Ollama, consumer GPU) over 60 hand-written questions on 12 RAG papers, with three replications per condition (66 runs total). Our primary contribution is characterizing this silent judge-overflow failure and a simple mitigation—capping judge context—and showing that the corrected effect is real but far smaller than the artifact suggested. We add two supporting audit lessons surfaced by the same ablation: a post-reranking sentence filter that removes answer-bearing evidence despite unchanged retrieval, and a document-level recall metric that masks a passage-level failure. Expected effects (larger generators help, rerankers help, smaller chunks help) behaved as background. This is a single-pipeline, single-author study, not a field-wide survey; the anonymized supplementary archive holds the full run grid and pre-/post-fix judge traces.

2 Experimental Setup

Corpus and questions. Our evaluation corpus comprises 12 recent RAG papers on query rewriting, self-RAG, context filtering, and related topics indexed into 558 word-based chunks (256 words, 50 overlap). We hand-wrote five questions per paper (60 total); `expected_source` is the gold PDF filename. This self-built benchmark risks convenience bias (Section 4); we use it to stress-test our own pipeline, not as a community benchmark.

Locked baseline. Hybrid BM25+dense retrieval (BGE-small; Xiao et al. 2024), BGE cross-encoder reranking, no context filter, Qwen2.5-14B generation under a strict JSON schema with deterministic citation parsing, and a separate Qwen2.5-14B judge. Final score is a weighted composite: 0.35

answer correctness + 0.25 faithfulness + 0.20 context recall + 0.10 context precision + 0.10 citation accuracy; the first four are LLM-judged, citation accuracy is deterministic. These weights are a pragmatic, untuned choice that foregrounds correctness and faithfulness; we report all five components separately (supplementary archive) so results can be reweighted, and keep deterministic citation accuracy at 0.10 as a lightweight grounding check rather than a primary signal. We additionally track Recall@5 (binary, *document*-level).

Ablation protocol. Each ablation changes one variable from the locked baseline with $n=3$ replications. The locked configuration scored 83.22 on the single development run that defined it; three fresh replications of that same configuration yield mean 81.79 ($\sigma=1.52$). We anchor all deltas to this three-replication mean rather than the single 83.22 point estimate—to avoid conflating one development run with the noise floor—and use $\sigma=1.52$ as a practical noise band, not a formal significance test (Section 4). Mid-study we discovered a judge failure (Section 3.1); we capped judge input at five chunks / 1,920 words while the generator still receives full top-k context, re-ran affected conditions, and archive pre-fix scores.

3 Findings

3.1 LLM-Judge Context Overflow

Ablation 8 (retrieval depth) and Ablation 9 (chunk size) initially appeared catastrophic. With top-$k=10$ or 512-word chunks, mean final scores were 9.9 and 14.1 (Table 1)—the kind of collapse that invites a reviewer to distrust the entire pipeline. Yet JSON parse rate stayed at 100% and deterministic citation accuracy remained near 80%; generation was not failing wholesale.

The judge, however, returned schema-valid JSON with all four LLM-scored fields set to zero when its input exceeded roughly 2,500 tokens in our setup ($\sim$2,560 words in our top-$k=10$ runs). Only citation accuracy ($\sim$10% of the composite) contributed, yielding scores of 8–14. Re-judging stored outputs with capped context immediately recovered per-question scores of 50–80. We implemented `JUDGE_MAX_CONTEXT_CHUNKS=5` and `JUDGE_MAX_CONTEXT_WORDS=1920` in our scoring module and re-ran both conditions. We have not yet run a controlled synthetic length sweep, so we treat $\sim$2,500 tokens as an observed onset in this pipeline rather than a model-intrinsic limit.

Condition	Pre-fix	Post-fix	Δ
A8 top-k=10	9.9	64.1 ± 2.1	-17.7
A9 chunk 512	14.1	65.9 ± 0.9	-15.9

Table 1: Judge overflow artifact vs. corrected scores (means over 3 runs; Δ vs. replicated baseline mean 81.8). Pre-fix archives retained for audit.

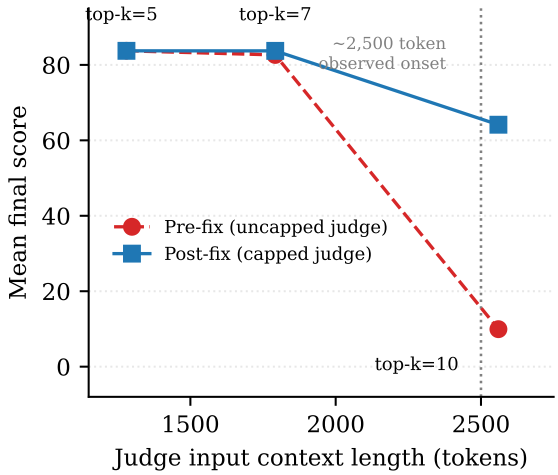

Figure 1: Mean final score vs. judge input context length (A8 top-k). Pre-fix scores collapse in the longest-context condition; the dotted line marks the observed onset in our setup. Post-fix capped judging reveals the remaining regression from context dilution at generation time.

Post-fix means are 64.1 ($\sigma=2.1$) for top-$k=10$ and 65.9 ($\sigma=0.9$) for chunk-512—still sizable regressions (-17.7 and -15.9 vs. the replicated baseline mean of 81.8), but now attributable to context dilution at generation time rather than judge failure. Top-$k \in \{5, 7\}$ and chunk sizes 128/256 remain within or just outside the noise band (Table 2). This failure mode is especially dangerous because silent zeros look like genuine poor grounding; nothing in a results table flags them as bugs. We tested one judge model, one local serving stack, and one prompt/schema; the transferable lesson is that RAG evaluations piping variable-length retrieved context into an LLM judge should audit for schema-valid degenerate outputs rather than trusting low scores at face value.

3.2 Audit Lesson 1: Post-Reranking Truncation Is Not Safe to Assume

Ablation 4 tests a heuristic sentence filter applied *after* cross-encoder reranking: keep the first 10 or first 5 sentences from each retrieved chunk, a fixed-rule stand-in for learned context filtering (Wang et al., 2023). We report it as an audit lesson, not a claim about context filtering in general: a fixed rule is a weak baseline for a trained selector, so the point is that post-reranking compression must be measured after reranking rather than assumed safe from retrieval metrics alone.

The fixed filter removes evidence the reranker had selected (Table 2). With no filter, mean final score is 82.9 (σ=1.0), within noise of baseline. Keeping the first 10 sentences costs 2.8 points (79.0); keeping only the first 5 costs **12.9 points** (68.9, σ=0.5)—the largest effect we observed among settings that do not change how many chunks reach the generator. Context recall drops from 86.4 to 62.8 (−23.6) while retrieval is unchanged; answer correctness falls from 79.4 to 63.5. As a concrete case, on `longrag_q2` ("the three main components of LongRAG") the reranker placed the gold chunk at rank 1 (score 0.925), but the sentence enumerating all three components (long retrieval unit, long retriever, long reader) was that chunk's sixth, so the first-five-sentence cap discarded it and per-question context recall fell from 90 to 0. Our own development trajectory had bundled this filter with other changes in an earlier run (67.1 → 83.2 when disabled); the ablation isolates the filter as the dominant regression.

Reranking surfaces high-scoring but sentence-sparse chunks; truncating to the first five sentences then removes the evidence the reranker selected. Learned filters may succeed where fixed heuristics fail, but the practical lesson is to evaluate post-reranking truncation directly rather than trusting it because upstream retrieval looks healthy.

3.3 Audit Lesson 2: Document Recall Masks Passage-Level Failure

We highlight one failure as a **case study**, not a corpus-wide prevalence claim. Question `adaptllm_q5` asks for the main bottleneck in ADAPT-LLM performance; the answer states that information retrieval is the bottleneck, supported by gold-vs.-retrieved passage comparisons. With hybrid retrieval and top-k=5, the generator receives five chunks; rank-1 is chunk 4 from the gold PDF—so document-level Recall@5 is **100%**—yet final score is **10.0** whenever the generator correctly refuses to answer from insufficient context.

Diagnosis: `pypdf` extraction splits the answer-bearing sentence across a chunk boundary.

Condition	**Mean**	**Δ**
Replicated baseline (3×)	81.8 ± 1.5	ref
Primary finding and audit lessons		
A8 top-k=10 (post-fix)	64.1 ± 2.1	−17.7
A9 chunk 512 (post-fix)	65.9 ± 0.9	−15.9
A4 filter first-5 sent.	68.9 ± 0.5	−12.9
`adaptllm_q5` Recall@5	100%	score 10.0
Background (expected)		
A2 Llama3.1-8B	73.7 ± 1.2	−8.1
A6 no reranker	78.1 ± 1.7	−3.7
A9 chunk 256	83.9 ± 0.8	+2.1

Table 2: Selected conditions (means ± SD over 3 runs; Δ vs. replicated baseline mean 81.8, σ=1.52). Remaining ablations fall within ∼2.5 points of baseline; A9 chunk-128 is −3.7.

Chunk 4 (passed to the generator as [Doc 1]) ends mid-word (*"even withou. . . "*) and omits the bottleneck conclusion; the completing span sits in chunk 5, which ranks outside the top-5 window. The model therefore sees a misleading partial passage from the correct file but not the answer-bearing span. The judge correctly scores the faithful refusal as wrong; an earlier 8B generator had scored 28–67 by hallucinating from the truncated chunk—higher scores, less grounding.

Document-level recall cannot distinguish retrieval of the right *file* from retrieval of the right *passage*; faithful RAG evaluation needs passage-level or answer-bearing-span metrics alongside aggregate means to catch this class of blind spot.

Other ablations (background). The remaining single-variable changes stay close to the replicated baseline (means of 81–84): retriever type (+0.4 hybrid, +0.3 dense), embedding size (+1.3 small, +2.4 large), output format (+1.8 JSON, +2.3 XML), and query transform (−0.4 HyDE) are all small relative to the headline regressions. Expected larger effects appear for generator size (−8.1, 8B vs. 14B), reranker removal (−3.7), and 128-word chunks (−3.7); we do not treat them as novel findings.

4 Conclusion

Faithful RAG evaluation requires checking not only whether models ground answers in retrieved text, but whether the evaluation stack itself fails silently. Our main finding is an LLM judge that returns schema-valid zeros past an observed onset around ∼2,500 tokens in our setup, turning a modest regression into an apparent collapse. Two sup-

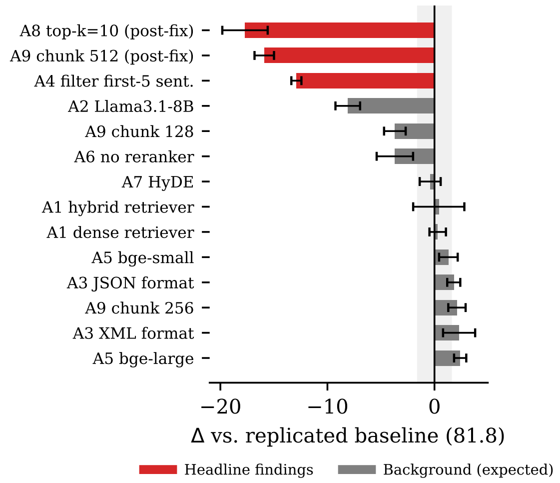

Figure 2: Δ from the replicated baseline mean (81.8) across conditions, ordered by magnitude. The primary finding and audit lessons (red) fall well outside the ± 1.52 noise band (shaded); background conditions (gray) behave as expected.

porting audit lessons—a post-reranking filter that removes grounding evidence despite perfect retrieval, and a document-level recall metric that certifies success while the answer-bearing passage is unavailable (the `adaptllm_q5` case study)—show related ways the stack can mislead. These issues surfaced only because we replicated each condition and audited anomalous scores rather than accepting them as pipeline failure. We release the experiment log (66 ablation runs, pre- and post-fix judge traces); an anonymized supplementary archive accompanies this submission, and a public repository will follow acceptance.

Limitations

Each condition uses n=3 replications; we report means and standard deviations but cannot claim statistical significance—only direction and magnitude. The 60 questions were written by the pipeline author over a self-selected paper set, a convenience benchmark we acknowledge openly. Our judge model changed during early development (Llama3.1-8B $\rightarrow$ Qwen2.5-14B); all reported ablations use the fixed later regime. Four of the five composite components are scored by the same judge whose silent failure we document; we have not validated post-fix scores against human labels, so a human-annotated correlation study is important future work before treating the corrected scores as ground truth. Findings describe one local pipeline and one academic-QA corpus; we hypothesize the *qualitative* patterns—silent judge failure, filter–reranker interaction, passage-level recall gaps—may recur elsewhere because they stem from common evaluation design choices rather than idiosyncratic model behavior, though exact point deltas may not transfer. No human subjects or sensitive data were involved.

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